3.2
Healthcare

The healthcare sector has emerged as a major driver of India's economy, significantly impacting both revenue and employment with public expenditure on healthcare being 2.1 % of GDP in FY23 as per the Economic Survey 2022-23. The healthcare industry includes diverse sub-sectors such as hospitals, medical devices, clinical trials, outsourcing, telemedicine, medical tourism, health insurance and medical equipment.

In 2023, the Indian healthcare industry witnessed substantial growth, with the sector valued at approximately US$230 billion. This growth is supported by advancements in healthcare infrastructure, increased digital health adoption, and rising private and public investments. The sector is expected to continue its expansion at a robust rate of 17-18% annually, reflecting its crucial role in India's economic development.

Employment within the sector records a significant contribution with over 5 million people employed in various healthcare roles as of 2023.

3.2.1
Macroeconomic trends

A Key statistics

The healthcare sector in India significantly contributes to the nation's economy and employment landscape.

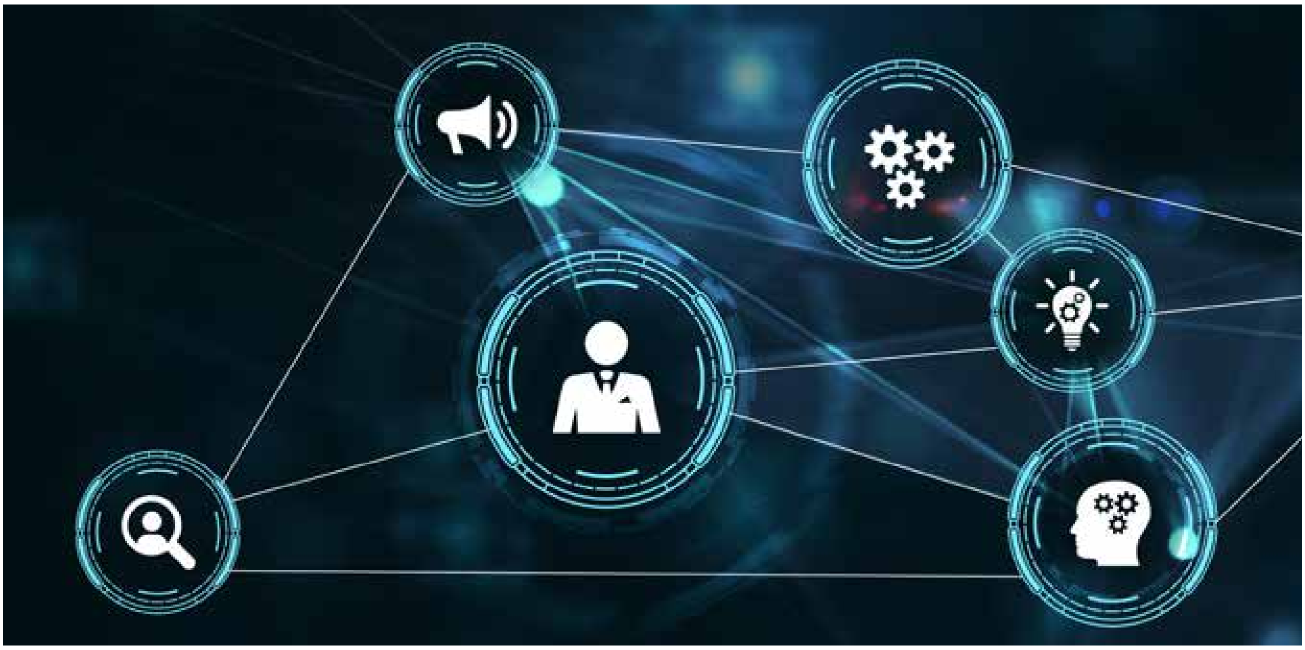
As of 2023,

the Indian healthcare market was valued at approximately
US$ 98.98 billion

Projections suggest a compound **annual growth rate (CAGR) of 8% from 2024 to 2032**, potentially reaching
US$ 638 billion by 2025[38]

The sector's contribution to India's GDP has been steadily increasing, with public expenditure on healthcare rising from
1.6% of GDP in FY21 to 2.2% in FY22
and it is projected to reach
2.5% by FY25

Conducive policies have encouraged FDI in the sector, for instance, from April 2000 to March 2024, the FDI inflow for the drugs and pharmaceuticals sector stood at US$22.57 billion followed by US$10.26 billion in hospitals and diagnostic centres, significantly enhancing the quality and accessibility of healthcare services nationwide[39].



B Employment scenario

As of 2024, the Indian healthcare sector ranks among the country's largest employers, providing jobs to a total of 7.5 million people. Advancements in telemedicine, virtual assistants and data analytics are anticipated to generate 2.7 to 3.5 million new jobs in technology-related roles[40]. This industry is poised for significant job growth owing to the expansion of healthcare facilities and rising demand for healthcare services. Between 2021 and 2023, there was a notable 22.4% increase in demand for healthcare professionals. This diverse workforce will include doctors, nurses, lab technicians, healthcare administrators, medical coders, AI engineers, pharmacists, dentists, data analysts and telemedicine specialists. This demand is expected to double by 2030 due to a shortage in this workforce.

C Key drivers of growth and government schemes

The 2024-25 Union Budget[41] allocated INR90, 659 crore to the Ministry of Health and Family Welfare, which is a 13% rise over the revised estimates of 2023-24. Key government initiatives driving this transformation:

- **Ayushman Bharat:** Launched in 2018, this health insurance scheme provides up to INR5 lakh (US$6,000) per family annually for hospitalizations. As of 2023, over 300 million Ayushman cards have been created, making it the largest health insurance program globally.[42]
- **National Digital Health Mission (NDHM):** The National Digital Health Mission (NDHM) launched in 2020 to digitize health records and integrate digital health services. By mid-2024, over 120 million individuals were registered, with the aim to cover more than 1.4 billion people.
- **The National Health Policy 2017 and Pradhan Mantri Jan Arogya Yojana (PMJAY)** have broadened healthcare coverage and improved service delivery, creating more job opportunities, especially in tier 2 and 3 cities. The National Healthcare Professional Registry and Accreditation System is crucial for managing the workforce by maintaining a comprehensive database of practitioners, ensuring regulatory compliance, and promoting continuous professional development. These policies help address immediate workforce shortages and support long-term planning.
- **The Indian Medical Association (IMA)** notes a continuing shortage of about 600,000 doctors as of 2024. Evolving technology and healthcare needs have given rise to an increasing demand for professionals skilled in the latest technologies and sustainable practices. Continuous upskilling is crucial to keep up with these changes and maintain a responsive, patient-focused healthcare system.

3.2.2 International benchmark

Healthcare Access and Quality (HAQ) Index is a measure developed to assess the performance of healthcare systems worldwide, focusing on how well a country or region provides access to and quality of healthcare services. The index ranges from 0 to 100, with higher scores indicating better access to and quality of healthcare.

> India ranks 44th in the global Healthcare Access and Quality (HAQ) Index with a score of 65.4. Key improvement areas include expanding infrastructure, increasing the workforce, and enhancing service quality and accessibility, particularly in rural areas.

Top-ranked countries are Taiwan (1st, score 89.5) for AI integration and quality services, South Korea (2nd, score 88.0) for sustainable practices and technology, and Japan (3rd, score 87.8) for personalized healthcare and prevention.

Worldwide, the USA and UK are leading the way in offering advanced programs that could benefit India. The USA's initiatives include cutting-edge training, focus on telemedicine, and AI diagnostics, while the UK's NHS focuses on comprehensive training and digital health tools. Adopting these practices could enhance India's healthcare system by improving access, quality, and efficiency.[43,44]

3.2.3 Sectoral trends

A Emerging technologies in healthcare sector

The top healthcare trends and innovations include Artificial Intelligence, the Internet of Medical Things, Telemedicine, Big Data and Analytics, Immersive Technology, Mobile Health, 3D Printing, Blockchain, Cloud Computing and Genomics[45].

According to the World Economic Forum, the healthcare sector is set for significant job growth in the coming years, especially in tech and data roles which will open new opportunities for professionals with healthcare knowledge with tech skills. The Indian healthcare sector is focusing on training its workforce in emerging technologies such as IoMT, blockchain, Telehealth, telemedicine Artificial Intelligence and Machine Learning, Big Data and Analytics and Augmented and Virtual Reality to enhance service delivery.

B Workforce trends

By 2030, the demand for healthcare professionals is expected to double due to a shortage of workers. Roles such as medical and health services managers, nurse practitioners, physician assistants, home health aides and physical therapists will see the highest growth[46].

Some of the other emerging job roles also highlighted during discussion in healthcare sector are as follows:

- **AI and Machine Learning specialists:** These professionals develop and implement AI algorithms for healthcare applications, such as diagnostic tools and predictive models.
- **Telemedicine coordinators:** They oversee virtual care programs, arrange remote consultations, and facilitate clear communication between patients and healthcare providers.
- **Health data scientists:** These professionals analyze complex healthcare data to extract insights, enhance patient outcomes and optimize healthcare operations.
- **IoMT systems engineers:** They design, implement, and maintain networks of connected medical devices and ensure data security and interoperability.
- **Digital health product managers:** Oversee the development and implementation of digital health solutions, ensuring they meet user needs and regulatory standards. They collaborate with stakeholders to drive product strategy, manage roadmaps, and optimize user experience.
- **Robot-assisted surgery technicians:** Surgical professionals assist surgeons in operating AI-powered robots and manage the technical aspects of robotic surgical systems during procedures.

3.2.4
Future of jobs: insights from industry leaders

A Emerging job roles in healthcare

The healthcare sector is poised for substantial transformation, leading to the creation of several new job roles. Our survey reveals the following job roles are anticipated to emerge in the near future:

Health data analyst: This role is expected to be in high demand, with 50% of respondents identifying it as a critical job role for the future. The increasing reliance on data-driven decision-making in healthcare underscores the need for professionals capable of analyzing vast datasets to improve patient outcomes and operational efficiency.

Genomic counsellor: Another emerging role, highlighted by 50% of respondents, is that of a genomic counsellor. As personalized medicine and genomics gain traction, the demand for specialized guidance based on genetic information will grow, making this role essential.

Digital health specialist: Identified by 33% of respondents, the Digital Health Specialist will play a key role in integrating and managing digital health technologies within healthcare systems. This position will be crucial as the sector continues to adopt more sophisticated digital tools.

Telemedicine coordinator: With 33% of respondents foreseeing the growth of telehealth services, the role of telemedicine coordinator is expected to become increasingly important for managing remote patient care.

AI healthcare ethicist: As AI integration in healthcare expands, 33% of respondents believed there will be a need for professionals who can navigate the ethical challenges associated with AI applications, ensuring that these technologies are used responsibly.

B Impact of Gen AI, AI, sector-specific technology, international mobility, DPI, climate, and green jobs on the job and skilling ecosystem

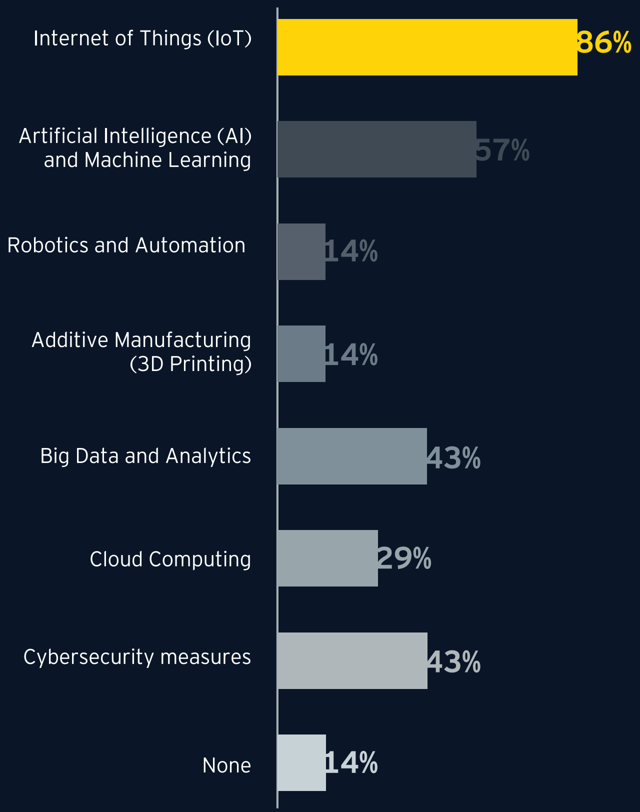
Artificial Intelligence (AI) and Gen AI: AI is expected to have a profound impact on the healthcare sector, with 66% of respondents rating its future impact as high as 9 or 10 on a scale of 10. However, respondents anticipate significant challenges in integration, including technical complexity, high costs of implementation and workforce resistance. 67% of the respondents expect AI to reshape job roles through reskilling and upskilling initiatives.

Sector-specific technology: Adoption of Industry 4.0 technologies is widespread, with 86% of respondents using IoT and 57% utilizing AI and Machine Learning. Despite this, the organizations feel there is lot of scope to be able to integrate AI and Gen AI, completely.

International mobility: Given that 71% of the organizations operate on an international scale, there is an increasing need for skills related to global operations. Cross-cultural competency and adaptability (50%) and global regulatory awareness (67%) are identified as crucial skills for the future healthcare workforce. Multilingual communication abilities are also highlighted by 33% of respondents as important for facilitating international mobility.

Climate and green jobs: Sustainability is a growing priority, with 43% of respondents identifying it as a top priority and another 43% considering it somewhat of a priority. However, 43% of organizations have not yet adopted any sustainable practices. Nevertheless, initiatives like waste reduction and recycling (67%) and increased energy efficiency (50%) are already underway in many organizations, highlighting a shift towards greener operations. On the jobs front, 67% of respondents affirm that economic feasibility and resistance to change are the primary challenges in creating and filling green jobs.

Utilization of any of the Industry 4.0 technologies



Challenges anticipated in integrating AI into existing Healthcare systems

50%
Technical Complexity

50%
High cost of implementation

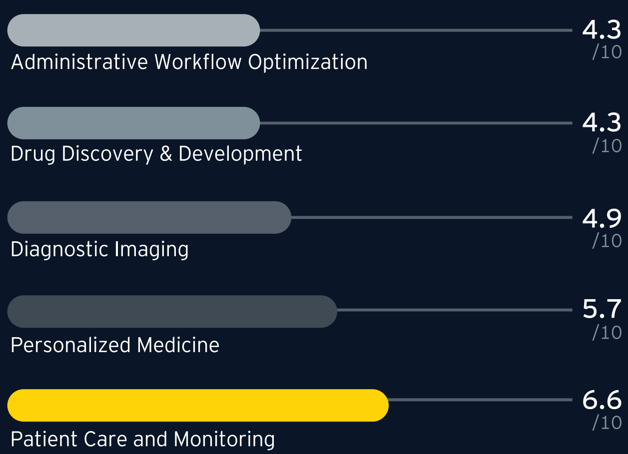
50%
Workforce resistance

33%
Lack of technical skills

83%
Data privacy and security concerns

How prepared are organizations to adapt AI and Gen AI

Expected Healthcare sectors to have greatest impact of AI



Suggested programmatic and policy interventions

For industry leaders

- **Reskilling and upskilling initiatives:** Industry leaders should prioritize regular training programs, as 67% of respondents emphasize the need for reskilling and upskilling to adapt to AI and other emerging technologies. These initiatives will be essential to address the gaps in technical skills and prepare the workforce for future challenges as industry 4.0 technologies take precedence.
- **Adopting sustainable practices:** With 43% of organizations already recognizing sustainability as a priority, it is critical to continue integrating sustainable practices into core operations. Green jobs, supported by targeted training programs, will be vital for this transition.
- **Enhanced collaboration with government:** To leverage government incentives effectively, industry leaders should engage in partnerships that align with policy objectives. These collaborations will help maximize the benefits of subsidies and incentives for technology adoption and sustainability.

How effective are current government policies in promoting technology adoption in the Healthcare sector

40%
Very effective

60%
Neutral

For government

- **Financial incentives for green investments:** With 60% of respondents indicating the need for financial incentives, the government should consider offering robust support to encourage the adoption of sustainable practices in healthcare.
- **Support for AI and technology integration:** The government can provide subsidies for employee training in new technologies, as 40% of respondents have identified this as beneficial. Stricter environmental regulations (40%) and incentives for adopting sustainable practices (60%) are also necessary to drive progress.
- **Policy support for workforce mobility and gender diversity:** To ensure an inclusive and diverse workforce, the government should consider implementing gender quotas (40%), offering grants for diversity initiatives (20%), and providing childcare support services (20%). Additionally, the government should play a supportive role (20%) in addressing skill gaps through public-private partnerships, ensuring that the healthcare sector is equipped to meet future demands.

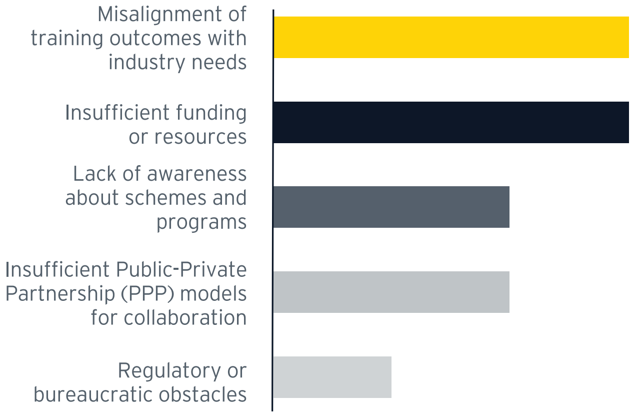
These interventions will be critical in shaping the future of jobs in the healthcare sector, ensuring that the workforce is equipped with the necessary skills to adapt to technological advancements, sustainability challenges, and global operations

Government incentives/subsidies can be/ have been most beneficial for technological advancements

40%	Subsidies for training employees in new technologies
60%	Incentives for adopting sustainable practices
20%	None have been beneficial

Areas skilling institutions should focus on

Challenges healthcare companies face in collaborating with educational and skilling institutions

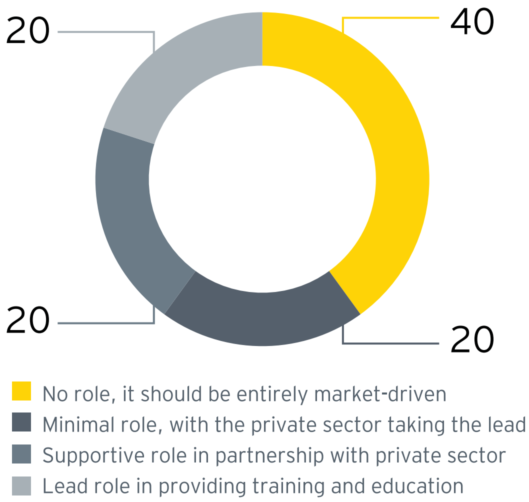


3.2.5
Overall impact on job roles and skills in the sector

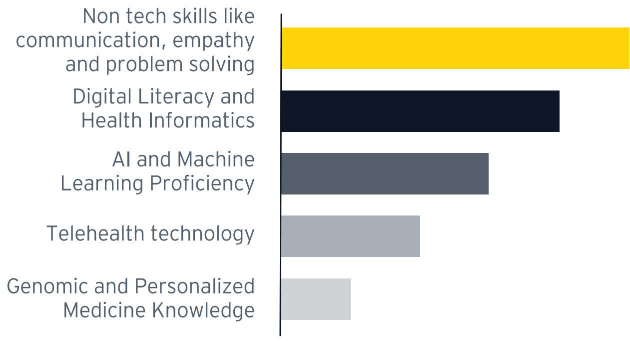
Our primary and secondary research in the healthcare field shows encouraging trends. Both types of research emphasize the importance of healthcare to India's economy. They consistently predict solid growth and the creation of new jobs due to technological advancements. Primary research highlights the emergence of new roles like Health Data Analysts, Genomic Counsellors, and Digital Health Specialists. According to 50% of respondents, these roles will be crucial in the future.

Emerging job roles in healthcare sector

Role should the government play in addressing skill gaps in the Healthcare sector



Skill sets that will be most in demand for future healthcare job roles



Full form of above mentioned variable "Non tech skills like communication , empathy and problem solving"

Collectively the research highlights the significant role of AI and emerging technologies in improving diagnosis and tailoring treatments. Research shows that 66% of respondents expect AI to significantly impact healthcare, although challenges such as technical complexity and resistance from the workforce remain. In the next few years, the healthcare sector will merge advanced technologies with traditional practices, creating exciting opportunities for those who can adapt and acquire the skills needed for the digital health era.